Algebraic Topology

I. Homotopy

1. Basic properties

~~Re~~ **Def I.1.1.** X & Y are spaces. $f_0, f_1 : X \to Y$ continuous then f_0 is homotopic to f_1

$(f_0 \cong f_1)$ iff $\exists$ continuous $F : X \times [0,1] \to Y$ s.t. $\begin{cases} F(x,0) = f_0(x) \\ F(x,1) = f_1(x) \end{cases}$ $\forall x \in X$.

Remark. (Gluing lemma) 1) X_i $(i \in I)$ subspaces. closed in X. $X = \bigcup_{i=1}^{n} X_n$. , $f_i : X_i \to Y$ continuous

Then $\exists f : X \to Y$ continuous. s.t. $f|_{X_1} = f_i$

2) X_i $(i \in I)$ subspaces open in X. $X = \bigcup_{i \in I} X_i$. $f_i : X_i \to Y$ continuous

Then $\exists f : X \to Y$ continuous s.t $f|_{X_i} = f_i$

Pf: $f : X \to Y$ is defined as $f(x) = f_i(x)$ iff $x \in X_i$ $\square$.

Thm I.1.2. Homotopy is the equivalence relation. on. the set of all continuous maps $X \to Y$.

Pf: ① Reflectivity, is obvious

② Symmetry: $f \cong g \Leftrightarrow \exists F : X \times [0,1] \to Y$, continuous. s.t. $F(x,0) = f(x)$. $F(x,1) = g(x)$.

let F' be $F'(x,t) = F(x, 1-t)$ ✓.

③ Transitivity: $f \cong g$, $g \cong h$, $\exists F, G$... let $H(x,t) := \begin{cases} F(x, 2t). & 0 \le t \le \frac{1}{2} \\ G(x, 2t-1) & \frac{1}{2} < t \le 1. \end{cases}$ $\square$.

Def I.1.3. $f : X \to Y$. continuous, its homotopy class is : $[f] = \{\text{continuous } g : g \cong f\}$.

Thm I.1.4. $f_i : X \to Y$. $g_i : Y \to Z$. $(i = 0, 1)$. If $f_0 \cong f_1$, $g_0 \cong g_1$ then $[g_0 \circ f_0] = [g_1 \circ f_1]$.

Pf: $\exists F : X \times [0,1] \to Y$. s.t. $F(x,0) = f_0(x)$, $F(x,1) = f_1(x)$

$G : Y \times [0,1] \to Z$. s.t $G(x,0) = g_0(x)$. $G(x,1) = g_1(x)$.

let. $H_1 : X \times [0,1] \to Z$. $H_1(x,t) = G(f_0(x), t)$.

then $H_1(x,0) = g_0 \circ f_0(x)$.

$H_1(x,1) = g_1 \circ f_0(x)$

thus $g_0 \circ f_0 \cong g_1 \circ f_0$

$H_2 : X \times [0,1] \to Z$. $H_2(x,t) = g_1(F(x,t))$ then $\begin{cases} H_2(x,0) = g_1 \circ f_0(x) \\ H_2(x,1) = g_1 \circ f_1(x) \end{cases}$

$\Rightarrow g_1 \circ f_0 \cong g_1 \circ f_1$.

thus $g_0 \circ f_0 \cong g_1 \circ f_1$ $\square$.

Def I.1.5 f $X \to Y$ is a homotopy equivalence iff $\exists g$ s.t $f \circ g \simeq 1_Y$ $g \circ f \simeq 1_X$.

X and Y have the same homotopy type iff $\exists f: X \to Y$ homotopy equivalence.

Def I.1.6. The constant map at y_0 is c $X \to Y$ $c(x) = y_0$ $\forall x \in X$.

$f: X \to Y$ is called nullhomotopic iff $\exists$ constant map c, s.t $f \simeq c$.

Thm I.1.7 $f: S^n \to Y$ continuous. then TFAE.

i) f is null homotopic.

ii) f can be extended to a continuous map: $D^{n+1} \to Y$.

iii) If $x_0 \in S^n$, $k: S^n \to Y$. is a constant map at $f(x_0)$.
then $\exists$ homotopic $F: f \simeq k$. s.t $F(x_0,t) = f(x_0)$. $\forall t \in [0,1]$.

Pf: i) $\Rightarrow$ ii): $F: f \simeq c$, $c(x) = y_0$ $\forall x \in S^n$

let $g: D^{n+1} \to Y$

$$g(x) = \begin{cases} y_0 & 0 \le \|x\| \le \frac{1}{2} \\ F\left(\frac{x}{\|x\|}, 2-2\|x\|\right) & \frac{1}{2} < \|x\| \le 1 \end{cases}$$

It's easy to check that g is continuous.

$\|x\| = 1$ 时 $g(x) = F(x,0) = f(x)$. $x \in S^n$

ii) $\Rightarrow$ (iii) let g be the extension of f.

Define $F: S^n \times [0,1] \to Y.$, $(x,t) \mapsto g((1-t)x + t x_0)$ is continuous

$$\Rightarrow \begin{cases} F(x,0) = g(x) = f(x) \quad \forall x \in S^n. \text{ thus } f \simeq k. \\ F(x,1) = g(x_0) = k(x) \end{cases}$$

~~then~~ and $F(x_0,t) = f(x_0)$ $\forall t \in [0,1]$.

(iii) $\Rightarrow$ (i) trivial. □.

2. Convexity, Contractibility and Cones.

Def I.2.1. A space X is contractible iff 1_X is null-homotopic.

Thm I.2.2. $X \subseteq \mathbb{R}^n$ is convex, then X is contractible.

Pf: let $x_0 \in X$. c is a constant map of x_0.

Define $F: X \times [0,1] \to X$ $F(x,t) = t x_0 + (1-t) x$

$\Rightarrow F(x,0) = x$. $F(x,1) = x_0 \Rightarrow 1_X \simeq c$ □.

Def I.2.2. (Cone). X is a space. Define an equivalence class by $\sim$ (where $(x,t) \sim (x',t')$ iff $t = t' = 1$). on $X \times [0,1]$. then the cone over $X := X \times I/\sim$ denoted by CX.

Thm I.2.3. The cone CX is always contractible.

Pf: Define $F: CX \times [0,1] \to CX$, $F([x,t],s) = [x, (1-s)t + st]$.

then $F([x,t],0) = [x,t]$. $F([x,t],1) = v$ □.

Next. we'll prove that controctible spaces are the simplest objects in hTop

Thm I.2.4. X has the same homotype as a point iff X is contractible

pf: $\Rightarrow$: Supposo X has the same homo-type as $\{a\}$.

$\exists\ f \cdot X \to \{a\}$ and $g : \{a\} \to X$. s.t. $g \circ f \simeq 1_X$. $f \circ g = 1_{\{a\}}$

$g(f(x)) = g(a) = x_0$ is constant $\Rightarrow$ X is contractible.

$\Leftarrow$: Suppose $1_X \simeq c$. $c :: X \to X$ $x \mapsto x_0$. Define $f : X \to \{x_0\}$

$g : \{x_0\} \to X$ $x_0 \mapsto x_0$ then $f \circ g = 1_{\{x_0\}}$. $g \circ f = c \simeq 1_X$. $\square$

3. Path Connectedness

Def. I.3.1 ①. the path components of X, is the equivalent class of X under $\sim$ ($a \sim b$ iff $\exists$ pa

from a to b)

② Define $\pi_0(X)$ be the set of all path components of X.

$f : X \to Y$. Define $\pi_0(f) : \pi_0(X) \to \pi_0(Y)$ $\pi_0(C) \mapsto$ the (unique) path component of Y

containing $f(C)$.

Thm I.3.2 $\pi_0 : \mathrm{Top} \to \mathrm{Sets}$ is a functor. Moreover, if $f \simeq g$. then $\pi_0(f) = \pi_0(g)$.

Pf: 1) It needs to show that $\forall\ f, g$ morphisms in Top. ($g \circ f$ is defined),

We have $\pi_0(g \circ f) = \pi_0(g) \circ \pi_0(f)$.

$f : X \to Y$. $g : Y \to Z$